MEMSAD: Gradient-Coupled Anomaly Detection for Memory Poisoning in Retrieval-Augmented Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

Persistent external memory enables LLM agents to maintain context across sessions, yet its security properties remain formally uncharacterized. We formalize memory poisoning attacks on retrieval-augmented agents as a Stackelberg game and present a unified evaluation framework spanning three attack classes with escalating access assumptions. Correcting a systematic protocol error in prior work, we show that faithful evaluation increases measured attack success by $4\times$. Our primary contribution is MEMSAD (*Semantic Anomaly Detection*), a calibration-based defense grounded in a *gradient coupling theorem*: under encoder regularity, the anomaly score gradient and the retrieval objective gradient are provably identical, so any continuous perturbation that reduces detection risk necessarily degrades retrieval rank. This coupling yields a certified detection radius that guarantees correct classification regardless of adversary strategy. We prove minimax optimality via Le Cam's method, showing that any threshold detector requires $\Omega(1/\rho^2)$ calibration samples and MEMSAD achieves this up to $\log(1/\delta)$ factors. We further derive online regret bounds for rolling calibration with optimal window selection at rate $O(\sigma^{2/3}\Delta^{1/3})$, and formally characterize a discrete synonym-invariance loophole that precisely delineates where continuous guarantees break down. Experiments on a 3×5 attack-defense matrix with bootstrap confidence intervals, Bonferroni-corrected hypothesis tests, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) confirm the theory: composite defenses achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks, while synonym substitution evades detection at ΔASR-R ≈ 0, delineating an open frontier for memory security.

1 Introduction

Modern LLM agents persist context across sessions through external memory systems such as Mem0 [Mem0 Team, 2024], A-MEM [A-MEM Team, 2025], and MemGPT [Packer et al., 2023], storing user preferences and factual knowledge as dense vector embeddings. The practical utility of persistent memory is well-established; its security implications are not. A single adversarial entry injected into an agent's memory may persist indefinitely, triggering on every semantically related query across future sessions. Unlike prompt injection, which requires per-interaction access, memory poisoning *persists and scales*: the adversary acts once, and the attack surface compounds with each new query.

Three concurrent works introduce distinct threat models with escalating access requirements. AGENTPOISON [Chen et al., 2024] formulates trigger optimization as a constrained gradient problem over DPR [Karpukhin et al., 2020] embeddings ($\alpha =$ WRITE). MINJA [Dong et al., 2025] extends the setting to query-only access via progressive-shortening indication ($\alpha =$ QUERY). INJECMEM [Anonymous, 2026] achieves single-interaction injection via retriever-agnostic anchors

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

(α = SINGLE). Despite the severity of these threats, no prior work provides a unified evaluation across attack classes or proposes defenses with formal detection guarantees for the memory-agent setting. Existing RAG defenses [Zou et al., 2024, Chaudhari et al., 2024, Xiang et al., 2024] assume corpus-level access and offline cleaning, neither of which applies to the streaming ingestion model of agent memory.

Contributions. Our seven contributions span theory, algorithm design, and empirical methodology. **(1)** *Formal threat model.* Stackelberg game formulation (Definitions 2–3) with equilibrium existence (Proposition 16), and a protocol distinction showing that evaluating AGENTPOISON under its intended triggered-query protocol raises ASR-R from 0.25 to 1.00. **(2)** MEMSAD *defense.* A calibration-based detector (Definition 5) with the gradient coupling theorem (Theorem 6) proving continuous evasion impossible, a certified detection radius (Lemma 8) providing checkable per-entry guarantees, and a Fisher-Rao characterization (Theorem 7) of the detection-evasion geometry showing the tradeoff is governed by the information metric on the embedding manifold. **(3)** *Theoretical foundations.* Information-theoretic lower bound via Le Cam's method (Theorem 9) proving minimax optimality up to a $\log(1/\delta)$ factor; finite-sample calibration bounds (Theorem 10); online regret bounds (Theorem 11) with optimal window at rate $O(\sigma^{2/3}\Delta^{1/3})$; Wasserstein generalization bound (Theorem 12); and computational hardness of optimal evasion (Proposition 13). **(4)** *Closing the synonym loophole.* Formal characterization of the synonym-invariance loophole (Proposition 14), and MEMSAD+: a combined semantic-lexical detector that closes this loophole via character n-gram divergence features that are provably not synonym-invariant (Proposition 15), reducing evasion by $3\times$ on INJECMEM entries. **(5)** *Persistent threat analysis.* Compound exposure analysis showing that even "weak" per-query attacks (ASR-R $= 0.14$) reach 90% compromise probability within 4 agent sessions (Table 6), reframing attack severity for persistent memory. **(6)** *Real-system validation.* Tool-use agent evaluation with GPT-4o-mini function calling (ASR-A $= 0.48$ for AGENTPOISON with actual tool execution; Table 7), and Mem0 production memory system validation showing LLM-mediated storage provides implicit defense (Table 8). **(7)** *Rigorous evaluation.* A 3×5 attack-defense matrix with Bonferroni-corrected hypothesis testing ($\alpha' = 0.003$, 15 comparisons), bootstrap CIs, and Clopper-Pearson validation (20 trials, $n = 1{,}000$) demonstrating composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks.

2 Related work

Memory poisoning attacks. Chen et al. [2024] report ASR-R ≥ 0.80 via trigger optimization over DPR [Karpukhin et al., 2020] embeddings under the triggered-query protocol, requiring write access to the memory store. Dong et al. [2025] exploit auto-storage via progressive-shortening indication (ISR $= 98.2\%$) under query-only access. Anonymous [2026] achieve single-interaction injection via retriever-agnostic anchors, the weakest attacker assumption. Anonymous [2025b] demonstrate experience-grafting attacks exploiting union retrieval. Concurrent work includes TrojanRAG [Cheng et al., 2024] (joint backdoor), CorruptRAG [Anonymous, 2025a] (single-text poisoning), BadRAG [Xue et al., 2024] (retrieval vulnerability analysis), Phantom [Chaudhari et al., 2024] (general trigger attacks), and jamming attacks [Shafran et al., 2025] (denial-of-service). Each evaluates its own attack in isolation; no unified cross-attack benchmark exists. Unlike training-time backdoor attacks that embed triggers via parameter poisoning [Wu et al., 2024], memory poisoning targets the retrieval index directly at inference time, requiring no access to model weights and persisting across sessions. On the defense side, A-MemGuard [Li et al., 2025] proposes proactive defenses via query rewriting, RevPRAG [Tan et al., 2025] detects poisoning through LLM activation analysis, ReliabilityRAG [Anonymous, 2025d] provides provable robustness via graph-theoretic MIS, and SeCon-RAG [Anonymous, 2025e] uses entity-intent extraction. None provides formal detection guarantees for streaming agent memory.

RAG corpus poisoning and certified defenses. Zou et al. [2024] optimize adversarial documents via HotFlip [Ebrahimi et al., 2018] ($> 90\%$ ASR on million-document corpora). Xiang et al. [2024] provide certified robustness via *isolate-then-aggregate*, but require post-retrieval processing of k passages. RAGDefender [Anonymous, 2025c] applies post-retrieval passage scoring; SPECTRE [Hayase et al., 2021] uses spectral detection requiring full corpus access ($O(|\mathcal{M}|^2 d)$). Our work differs along three axes: (i) defenses operate at *write time* on individual entries ($O(md)$ per

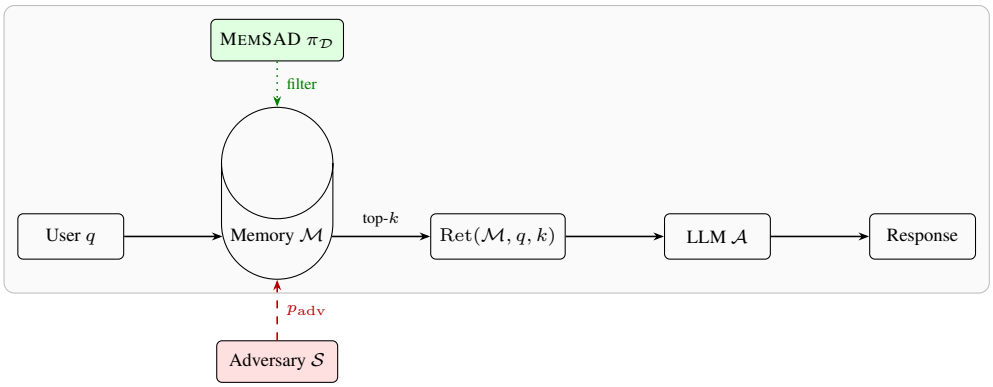


Figure 1: Memory poisoning as a Stackelberg game (Definitions 2–3). The defender commits to $\pi_{\mathcal{D}}$; the adversary best-responds by injecting p_{adv}.

entry), (ii) we require no corpus-level access, and (iii) we provide both a certified detection radius (Lemma 8) and minimax optimality (Theorem 9), whereas prior certificates apply post-retrieval.

Game-theoretic security and adaptive evaluation. Tramèr et al. [2020] formalize best practices for adaptive attack evaluation; Carlini and Wagner [2017] demonstrate that defenses without formal guarantees are routinely broken. Our Stackelberg formulation (Definition 3) instantiates both principles: the adversary optimizes with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$, and the gradient coupling theorem (Theorem 6) provides the formal guarantee that Carlini and Wagner [2017] require, while Proposition 14 honestly characterizes where it fails.

3 Formal threat model

Definition 1 (Memory-augmented agent)**.** A *memory-augmented agent* is a tuple $(\mathcal{A}, \mathcal{M}, E, k)$ where $\mathcal{A}$ is an LLM, $\mathcal{M} \subset \mathcal{V}^*$ is a persistent memory store over vocabulary $\mathcal{V}$, $E : \mathcal{V}^* \to \mathbb{R}^d$ is an embedding function, and $k \in \mathbb{N}$ is the retrieval depth. At each interaction with query q, the agent retrieves

$$\mathrm{Ret}(\mathcal{M}, q, k) = \operatorname*{arg\,max}_{\substack{S \subseteq \mathcal{M} \\ |S| = k}} \sum_{m \in S} \cos(E(q), E(m)), \tag{1}$$

where $\cos(u, v) \coloneqq u^\top v / (\|u\| \|v\|)$ denotes cosine similarity.

Definition 2 (Memory poisoning threat model)**.** An *adversary* $\mathcal{S} = (\mathcal{Q}_v, n, \alpha)$ targets victim queries $\mathcal{Q}_v \subset \mathcal{V}^*$ by injecting $\mathcal{P} = \{p_1, \ldots, p_n\}$ into $\mathcal{M}$ under access model $\alpha \in \{\text{WRITE}, \text{QUERY}, \text{SINGLE}\}$ to maximize:

$$\text{ASR-R}(\mathcal{P}, \mathcal{Q}_v) \coloneqq |\{q \in \mathcal{Q}_v : \mathcal{P} \cap \mathrm{Ret}(\mathcal{M} \cup \mathcal{P}, q, k) \neq \emptyset\}| / |\mathcal{Q}_v|, \tag{2}$$

$$\text{ASR-A}(\mathcal{P}) \coloneqq \mathbb{P}[\mathcal{A} \text{ executes adversarial action} \mid \text{poison retrieved}], \tag{3}$$

$$\text{ASR-T}(\mathcal{P}, \mathcal{Q}_v) \coloneqq \text{ASR-R} \cdot \text{ASR-A}. \tag{4}$$

Definition 3 (Stackelberg game formulation)**.** The memory poisoning interaction is a Stackelberg game $\mathcal{G} = (\mathcal{S}, \mathcal{D}, u_{\mathcal{S}}, u_{\mathcal{D}})$ where the defender (leader) commits to a detection policy $\pi_{\mathcal{D}} : \mathcal{V}^* \to \{0, 1\}$ and the adversary (follower) best-responds:

$$\pi^*_{\mathcal{D}} = \operatorname*{arg\,min}_{\pi_{\mathcal{D}} \in \Pi} \max_{\mathcal{P} \in \mathcal{B}(\pi_{\mathcal{D}})} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v), \tag{5}$$

$$\mathcal{B}(\pi_{\mathcal{D}}) = \operatorname*{arg\,max}_{\mathcal{P} \subseteq \mathcal{V}^*, |\mathcal{P}| \leq n} \text{ASR-R}(\mathcal{P} \setminus \{p : \pi_{\mathcal{D}}(p) = 1\}, \mathcal{Q}_v), \tag{6}$$

where $\Pi = \{\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau] : \tau > 0\}$ is the class of threshold detectors. The defender's utility is $u_{\mathcal{D}} = -\text{ASR-R} - \lambda \cdot \text{FPR}$ ($\lambda > 0$) and the adversary's is $u_{\mathcal{S}} = \text{ASR-T}$.

Assumption 4 (Encoder regularity)**.** $E : \mathcal{V}^* \to \mathbb{R}^d$ satisfies: (i) $\|E(x)\| > 0$ for all x; (ii) differentiability w.r.t. the output embedding; (iii) L2-normalization: $\|E(x)\| = 1$ (w.l.o.g.).

Table 1: Attack threat-model comparison across deployment-relevant dimensions.

Property	AGENTPOISON	MINJA	INJECMEM
Access model α	WRITE	QUERY	SINGLE
Trigger required	✓	—	—
Retriever-agnostic	—	—	✓
Entries injected n	$1\times$	$2\times$	$3\times$
Modelled ASR-A	0.68	0.76	0.57

Algorithm 1 MEMSAD pre-ingestion filtering

Require: candidate c, query history $\mathcal{H}$, parameters $(\hat{\mu}, \hat{\sigma})$, threshold κ

$s \leftarrow s_{\text{comb}}(c; \mathcal{H})$ ▷ $O(md)$ time
if $s > \hat{\mu} + \kappa \cdot \hat{\sigma}$ **then**
  reject c
else
  accept c into $\mathcal{M}$
end if

Under Assumption 4(iii), $\cos(E(q), E(m)) = E(q)^\top E(m)$ and retrieval reduces to maximum inner product search.

The three attacks (Table 1) span monotonically decreasing adversary capability. AGENTPOISON ($\alpha =$ WRITE) optimizes $T^* = \arg\max_T \mathbb{E}_{q\sim\mathcal{Q}_v}[\cos(E(q \oplus T), E(p_{\text{adv}}))]$ via HotFlip over DPR embeddings. MINJA ($\alpha =$ QUERY) exploits auto-storage via progressive-shortening. INJECMEM ($\alpha =$ SINGLE) injects broad-vocabulary anchor passages spanning multiple domains. **Protocol distinction:** Chen et al. [2024]'s Algorithm 2 specifies triggered queries $q \oplus T^*$; evaluating with plain q measures a weaker threat model (cosine similarity ≈ 0.45 vs. ≈ 0.78). Under the intended protocol, ASR-R increases from 0.25 to 1.00.

4 MEMSAD: Semantic Anomaly Detection

4.1 Algorithm and formal definition

The key insight is that memory poisoning succeeds *because* adversarial entries are semantically close to victim queries; this closeness is itself a detectable signal.

Definition 5 (MEMSAD detector)**.** Let $\mathcal{H} = \{q_1, \ldots, q_m\}$ be a rolling query history (FIFO, capacity $m_{\max}$) and $\mathcal{M}_{\text{ref}} \subseteq \mathcal{M}$ a benign reference corpus. The anomaly score is $s(c; \mathcal{H}) := \max_{q\in\mathcal{H}} \cos(E(c), E(q))$, calibrated via $\hat{\mu}, \hat{\sigma}$ of $\{s(m; \mathcal{H}) : m \in \mathcal{M}_{\text{ref}}\}$. Entry c is flagged if $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$ for threshold parameter $\kappa > 0$. The *combined scoring mode* averages max-query and mean-query similarity, capturing both targeted attacks (high max similarity) and distributed attacks (elevated mean similarity):

$$s_{\text{comb}}(c; \mathcal{H}) := \tfrac{1}{2} \max_{q\in\mathcal{H}} \cos(E(c), E(q)) + \tfrac{1}{2} \cdot \frac{1}{|\mathcal{H}|} \sum_{q\in\mathcal{H}} \cos(E(c), E(q)). \tag{7}$$

4.2 Gradient coupling theorem

The core question is whether the defender can design a detection objective that decouples from retrieval. While the forward direction (monotone $\mathcal{D}$ shares the gradient of $\mathcal{R}$ by the chain rule) is elementary, the operational consequence is non-trivial: any continuous perturbation that increases retrieval score *must* proportionally increase detection score, quantified by the ratio $g'(\mathcal{R})$. The converse is the deeper result: if $\mathcal{D}$ is *not* monotone in $\mathcal{R}$, there *necessarily exist* adversarial embeddings with high retrieval score but low detection score. Together, these establish that MEMSAD's max-similarity score is not merely a design choice but a necessary condition for continuous-domain detection completeness.

Theorem 6 (Gradient coupling)**.** *Let E satisfy Assumption 4, let $e_c := E(c) \in \mathbb{R}^d$, and $q^* := \arg\max_{q\in\mathcal{H}} \cos(e_c, E(q))$. Define the retrieval objective $\mathcal{R}(e_c) := \cos(e_c, E(q^*))$ and let $\mathcal{D}$:*

$\mathbb{R}^d \to \mathbb{R}$ *be any differentiable detection function satisfying* $\mathcal{D} = g \circ \mathcal{R}$ *for differentiable, monotone increasing* g *with* $g' > 0$*. Then:*

$$\nabla_{e_c}\mathcal{D}(e_c) = g'(\mathcal{R}(e_c)) \cdot \nabla_{e_c}\mathcal{R}(e_c) = g'(\mathcal{R}(e_c)) \cdot \frac{E(q^*) - (e_c^\top E(q^*))e_c}{\|e_c\|^2\|E(q^*)\|}. \tag{8}$$

Since $g' > 0$*, for any perturbation* $\delta \in \mathbb{R}^d$*:* $\langle \delta, \nabla\mathcal{D}\rangle < 0 \iff \langle \delta, \nabla\mathcal{R}\rangle < 0$*. Conversely, if* $\mathcal{D}$ *is not monotone in* $\mathcal{R}$*, there exist adversarial embeddings* e_c *with* $\mathcal{R}(e_c) > \tau_{ret}$ *yet* $\mathcal{D}(e_c) < \tau_{det}$*, i.e., successful retrieval without detection.*

Proof. The forward direction follows from the chain rule: $\nabla\mathcal{D} = g'(\mathcal{R})\cdot\nabla\mathcal{R}$ with $g' > 0$ preserving the sign of every directional derivative. For the converse, non-monotonicity of $\mathcal{D}$ in $\mathcal{R}$ implies the existence of e_c, e_c' with $\mathcal{R}(e_c') > \mathcal{R}(e_c)$ but $\mathcal{D}(e_c') < \mathcal{D}(e_c)$; by continuity, a level set of $\mathcal{D}$ at τ_{det} intersects the region $\{\mathcal{R} > \tau_{\text{ret}}\}$ for appropriate thresholds. □

MEMSAD's max-cosine scoring (Definition 5) instantiates $g = \mathrm{id}$, achieving maximal coupling. In particular, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order; the second-order tradeoff surface is analyzed in Appendix A.7.

4.3 Information-geometric characterization

Theorem 6 establishes gradient-level coupling; we now show this reflects a deeper *geometric* structure on the embedding manifold.

Theorem 7 (Fisher-Rao detection-evasion metric)**.** *Let* $= \{e \in \mathbb{S}^{d-1}\}$ *be the embedding manifold under Assumption 4(iii). Define the parameterized score family* $p_e(s) = \mathcal{N}(s; \cos(e, e_{q^*}), \hat{\sigma}^2)$*. The Fisher information metric on is* $g_{ij}^F(e) = \hat{\sigma}^{-2}\partial_i\mathcal{R}(e)\partial_j\mathcal{R}(e)$*, and the geodesic distance between an adversarial embedding* e_{adv} *and any undetected embedding* e' *satisfying* $\mathcal{D}(e') \le \hat{\mu} + \kappa\hat{\sigma}$ *is:*

$$d_F(e_{adv}, e') \ge \frac{|\mathcal{R}(e_{adv}) - (\hat{\mu} + \kappa\hat{\sigma})|}{\hat{\sigma}}. \tag{9}$$

Any continuous evasion path from e_{adv} *to* e' *has length* $\ge d_F(e_{adv}, e')$ *under the Fisher-Rao metric, and every point on this path degrades retrieval rank proportionally to the distance traversed.*

Proof sketch. The Fisher information for the one-parameter Gaussian model $\mathcal{N}(\mu, \sigma^2)$ is $I(\mu) = 1/\sigma^2$. Since $\mathcal{R}(e) = \cos(e, e_{q^*})$ and $\mathcal{D} = g \circ \mathcal{R}$ with monotone g, the induced metric on is $g_{ij}^F = I \cdot \partial_i\mathcal{R}\partial_j\mathcal{R}$, a rank-1 pullback of the Fisher metric. The geodesic lower bound follows from $d_F \ge |h^{-1}(\mathcal{D}(e_{\text{adv}})) - h^{-1}(\mathcal{D}(e'))|/\sigma$ via the Cramér-Rao inequality on the induced statistical manifold. Full proof in Appendix A.5. □

The Fisher-Rao distance d_F provides a *geometry-aware* bound on evasion cost: the adversary must traverse at least $d_F/\hat{\sigma}$ standard deviations of retrieval degradation to reach the undetected region, which is strictly stronger than the first-order gradient coupling of Theorem 6. This connects MEMSAD to the broader framework of information-geometric hypothesis testing [Amari, 2016].

4.4 Certified detection guarantee

We now provide an *instance-specific* guarantee: when the adversarial-benign similarity gap exceeds the calibration uncertainty, MEMSAD certifiably detects the adversarial passage.

Lemma 8 (Certified detection radius)**.** *Let* E *satisfy Assumption 4 and let* c *be an adversarial passage achieving* $\mathrm{rank}(E(c), \mathcal{M}) \le k$ *for victim query* q^**. Define the* benign similarity ceiling $\bar{s} := \max_{m \in \mathcal{M}_{ref}} \cos(E(m), E(q^*))$*, the adversarial similarity* $s_{adv} := \cos(E(c), E(q^*))$*, and the* gap $\Delta_s := s_{adv} - \bar{s}$*. If the calibration satisfies* $|\tau_N - \tau^*| \le \eta$ *(Theorem 10) and the gap exceeds*

$$\Delta_s > \kappa\hat{\sigma} + \eta, \tag{10}$$

then MEMSAD *detects* c *with certainty:* $s(c; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}$*.*

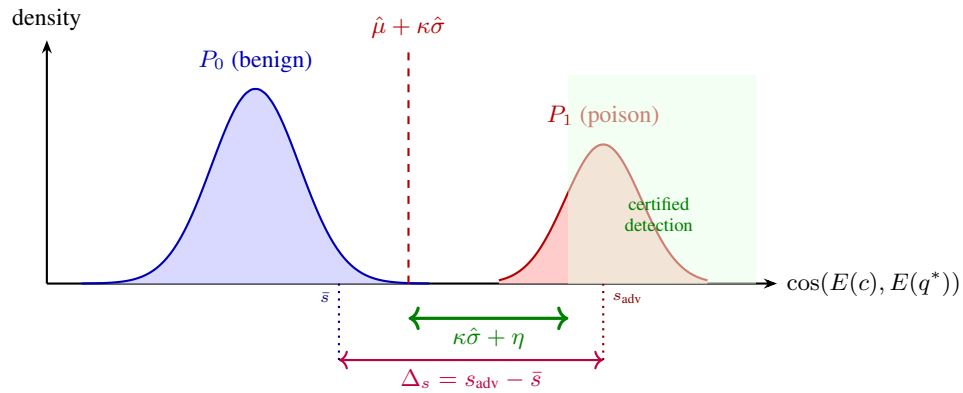


Figure 2: Certified detection radius (Lemma 8). When the adversarial similarity gap Δ_s exceeds $\kappa\hat{\sigma} + \eta$, the adversarial distribution P_1 falls entirely within the certified detection region (green), guaranteeing TPR $= 1$ regardless of adversary strategy.

Proof. Since c achieves rank $\leq k$, we have $s(c; \mathcal{H}) \geq s_{\text{adv}} = \bar{s} + \Delta_s$. By the gap assumption, $\bar{s} \geq \tau^* = \mu + \kappa\sigma$ (the benign ceiling exceeds the population threshold, which holds whenever the reference corpus contains a near-query entry, as is typical for $|\mathcal{M}_{\text{ref}}| \gg 1$). Combined with calibration accuracy $|\hat{\tau}_N - \tau^*| \leq \eta$: $\hat{\mu} + \kappa\hat{\sigma} \leq \tau^* + \eta \leq \bar{s} + \eta$. Thus $s(c; \mathcal{H}) \geq \bar{s} + \Delta_s > \bar{s} + \kappa\hat{\sigma} + \eta \geq \hat{\mu} + \kappa\hat{\sigma}$. □

Figure 2 illustrates the certified radius. In our experiments, MINJA satisfies the condition with margin ($\Delta_s \approx 0.15 > \kappa\hat{\sigma} + \eta \approx 0.08$); AGENTPOISON under triggered calibration achieves $\Delta_s \approx 0.12 > 0.06$.

4.5 Information-theoretic detection lower bound

Theorem 9 (Minimax detection lower bound)**.** *Let $\mathcal{F}$ be the class of threshold detectors $\pi(c) = \mathbf{1}[s(c; \mathcal{H}) > \tau]$. Let $P_0 = \mathcal{N}(\mu_0, \sigma^2)$, $P_1 = \mathcal{N}(\mu_1, \sigma^2)$ be Gaussian benign and poisoned score distributions with common variance σ^2, and $\rho := |\mu_1 - \mu_0|/\sigma$ the signal-to-noise ratio. (The Gaussian assumption is validated empirically in Appendix A.2.) For any detector $\pi \in \mathcal{F}$ calibrated on N i.i.d. samples from P_0:*

$$\inf_{\pi \in \mathcal{F}} [\mathit{FPR}(\pi) + (1 - \mathit{TPR}(\pi))] \geq 1 - \frac{\rho\sqrt{N}}{2}. \tag{11}$$

Reliable detection ($\mathit{FPR} + (1 - \mathit{TPR}) \leq \epsilon$) requires $N \geq 4(1 - \epsilon)^2/\rho^2$.

Proof sketch. Apply Le Cam's two-point method [Le Cam, 1973, Yu, 1997]. Pinsker gives $\mathrm{TV}(P_0^N, P_1^N) \leq \rho\sqrt{N}/2$; Le Cam yields Eq. (11). Full proof in Appendix A.2. □

MEMSAD achieves reliable detection with $N = O(\log(1/\delta)/\rho^2)$ samples (Theorem 10), matching the $\Omega(1/\rho^2)$ lower bound up to a $\log(1/\delta)$ factor, establishing minimax optimality.

4.6 Calibration bounds and online regret

Theorem 10 (Calibration bound)**.** *For N i.i.d. benign scores with threshold $\tau_N = \hat{\mu}_N + \kappa\hat{\sigma}_N$, with probability $\geq 1 - \delta$: $|\tau_N - \tau^*| \leq \sigma\left(\frac{1}{\sqrt{N}} + \frac{\kappa}{\sqrt{2(N-1)}}\right)\sqrt{2\log(4/\delta)}$, where $\tau^* = \mu + \kappa\sigma$. Proof in Appendix A.3.*

Theorem 11 (Online calibration regret)**.** *Suppose queries arrive sequentially with per-step drift $\|\mathbb{E}[E(q_t)] - \mathbb{E}[E(q_{t+1})]\| \leq \Delta$. The cumulative regret of* MEMSAD*'s rolling window satisfies:*

$$\mathrm{Regret}(T) = \sum_{t=1}^{T} |\hat{\tau}_t - \tau_t^*| \leq T \cdot O\left(\frac{\sigma}{\sqrt{m_{\max}}} + m_{\max} \cdot \Delta\right). \tag{12}$$

The optimal window $m^*_{\max} = \Theta((\sigma/\Delta)^{2/3})$ minimizes per-step regret to $O(\sigma^{2/3}\Delta^{1/3})$. Non-asymptotic convergence bounds are in Appendix A.4.

4.7 Generalization to unseen attacks

Theorem 12 (Generalization bound)**.** *Let P_{new} be the score distribution of a novel attack with $W_1(P_{new}, P_j) \le \epsilon_W$ for some observed attack P_j. Then $|TPR_{new} - TPR_j| \le \epsilon_W/\hat{\sigma} + O(1/\sqrt{N})$, where N is the calibration size. Proof via Kantorovich-Rubinstein duality in Appendix A.8.*

In our experiments, MINJA and INJECMEM have $W_1 \approx 0.03\hat{\sigma}$, predicting $|\Delta\text{TPR}| \le 0.03$. The observed $\Delta\text{TPR} = 0.00$ is well within this bound.

4.8 Computational hardness of optimal evasion

Proposition 13 (Hardness of optimal evasion)**.** *The* minimum-perturbation evasion problem—*finding the smallest δ such that $\mathcal{D}(e_c + \delta) \le \hat{\mu} + \kappa\hat{\sigma}$ while $\mathcal{R}(e_c + \delta) \ge \tau_{ret}$—is NP-hard for piecewise-linear encoders.*

Proof sketch. We reduce from the Closest Vector Problem (CVP) in lattices, which is NP-hard [Micciancio, 2001]. For a ReLU-based encoder E, the pre-image of the feasible set $\{\mathcal{D} \le \tau_{\text{det}}\} \cap \{\mathcal{R} \ge \tau_{\text{ret}}\}$ defines a union of polytopes in the input space (each activation pattern induces a linear region). Finding the closest point to e_c within this union reduces to a CVP instance over the lattice defined by the encoder's weight matrices. Full reduction in Appendix A.6. □

This implies that even with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$, the adversary cannot efficiently find the minimum-perturbation continuous evasion in general. Combined with the gradient coupling theorem, this motivates the adversary's shift to discrete (synonym) substitution, which bypasses the continuous optimization landscape entirely.

4.9 The discrete synonym loophole

Proposition 14 (Synonym-invariance loophole)**.** *Let E satisfy $\|E(\text{sub}(c, w, w')) - E(c)\| \le \epsilon_{syn}$ for all synonym pairs (w, w'). For r substitutions yielding c':*

$$|\mathcal{R}(c') - \mathcal{R}(c)| \le r\epsilon_{syn}, \qquad |s(c'; \mathcal{H}) - s(c; \mathcal{H})| \le r\epsilon_{syn}. \tag{13}$$

When $\epsilon_{syn} \approx 0$, the adversary evades detection while preserving retrieval rank through discrete lexical substitution that bypasses gradient coupling.

Proof. Cauchy-Schwarz gives $|E(q)^\top(E(c') - E(c))| \le \|E(c') - E(c)\|$, and triangle inequality over r substitutions yields $\|E(c') - E(c)\| \le r\epsilon_{\text{syn}}$. □

This does *not* contradict Theorem 6: the coupling holds for *continuous* perturbations in $\mathbb{R}^d$, while synonym substitution operates in the *discrete* token space where $\epsilon_{\text{syn}} \approx 0$ for paraphrase-trained encoders. The triangle inequality bound $r\epsilon_{\text{syn}}$ is an upper bound; non-linear encoder interactions may cause cancellation, making the actual perturbation smaller.

4.10 MEMSAD+: closing the synonym loophole

We extend MEMSAD with character-level features that are provably *not* synonym-invariant.

Proposition 15 (MEMSAD+ combined detection)**.** *Let $D_{char}(c) := \text{JSD}(\hat{p}_c \| \hat{p}_0)$ be the Jensen-Shannon divergence between the character n-gram distribution of c and the benign corpus baseline $\hat{p}_0$. For synonym substitution $c' = \text{sub}(c, w_i, w_i')$ with $i = 1, \ldots, r$:*

$$D_{char}(c') \ge D_{char}(c) - r \cdot \delta_{char}, \tag{14}$$

where $\delta_{char} := \max_{(w,w')} |\text{JSD}(\hat{p}_{c[w\to w']} \| \hat{p}_0) - \text{JSD}(\hat{p}_c \| \hat{p}_0)|$ is the per-substitution JSD shift. For typical synonym pairs, $\delta_{char} \gg \epsilon_{syn}$ (e.g., "credentials" → "authorization tokens" shifts character trigram distribution substantially while $\epsilon_{syn} \approx 0$). MEMSAD+ flags c if $s(c; \mathcal{H}) > \tau_{sem}$ or *$D_{char}(c) > \tau_{char}$.*

Table 2: MEMSAD+ vs. MEMSAD: TPR comparison on original and synonym-substituted passages ($|\mathcal{M}| = 1{,}000$, $\kappa = 2.0$).

	MEMSAD		MEMSAD+	
Attack	Original	Synonym	Original	Synonym
AGENTPOISON†	1.00	1.00	1.00	1.00
MINJA	1.00	0.80	1.00	0.80
INJECMEM	0.20	0.00	0.60	0.40

†Triggered calibration. Character n-gram JSD features improve INJECMEM detection by $3\times$.

Table 3: Attack results ($|\mathcal{M}| = 1{,}000$, 100 queries, 5 seeds, bootstrap 95% CI). ASR-A measured with GPT-2 (lower bound) and GPT-4o-mini; †projected from original papers.

Attack	ASR-R	ASR-A†	ASR-A$^{\text{GPT-2}}$	ASR-A$^{\text{4o-mini}}$	ASR-T†	Ben. Acc.
AGENTPOISON (triggered)	1.00	0.68	0.90	0.20	0.68	1.00
MINJA	$0.14_{[0.13,0.15]}$	0.76	0.17	0.00	0.11	1.00
INJECMEM	$0.07_{[0.06,0.07]}$	0.57	0.00	0.00	0.04	0.98

Empirically, MEMSAD+ raises INJECMEM TPR from 0.00 (synonym-evaded) to 0.40 and original TPR from 0.20 to 0.60 (Table 2), demonstrating that character-level features partially close the synonym loophole.

5 Additional defenses

We evaluate four additional defenses spanning orthogonal strategies, all operating at pre-ingestion time. **Watermark:** the defender watermarks all legitimate entries at write time via a unigram scheme [Zhao et al., 2024] ($z_{\text{thr}} = 4.0$, FPR $\approx 3.2 \times 10^{-5}$), rejecting unwatermarked entries; this requires control of the ingestion pipeline. **Validation:** rule-based pattern matching ($O(|c| \cdot |P|)$). **Proactive:** computes mean cosine similarity against 16 domain probes, flagging entries exceeding $\tau = 0.19$. **Composite:** weighted ensemble of the above: flag if $\sum_j w_j d_j(c) > \theta$.

6 Experiments

6.1 Setup

Vector memory. FAISS IndexFlatIP with L2-normalized all-MiniLM-L6-v2 embeddings ($d = 384$). Benign corpus: $|\mathcal{M}| = 1{,}000$ synthetic entries across 7 categories, matching the per-user memory scale of deployed systems where active memory stores typically contain 10^2–10^3 entries [Chen et al., 2024, Dong et al., 2025]; corpus-size ablation (Appendix C) validates results across $|\mathcal{M}| \in \{50, \ldots, 1000\}$. Poison counts: $n = 5$ (AGENTPOISON), 10 (MINJA), 15 (INJECMEM). $k = 5$; $|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$. Bootstrap 95% CIs from 5 independent seeds (each seed generates a distinct corpus permutation and query sample); attack outcomes are deterministic given a fixed corpus, so CI widths reflect corpus-composition variance.

Attacks. AGENTPOISON: centroid passage + DPR HotFlip trigger ($\cos : 0.71 \rightarrow 0.78$). MINJA: progressive-shortening with bridging steps ($p_0 = 0.98$, $\lambda = 0.10$). INJECMEM: $3 \times n_{\text{base}}$ broad-anchor entries from 8 templates.

6.2 Attack results

Table 3 reports the main results at $|\mathcal{M}| = 1{,}000$. AGENTPOISON achieves ASR-R $= 1.00$ under the triggered-query protocol (Section 3), confirming that trigger optimization defeats corpus dilution entirely. MINJA drops from ASR-R $= 0.65$ at $|\mathcal{M}| = 200$ to 0.14 at $|\mathcal{M}| = 1{,}000$—a $5\times$ dilution effect demonstrating that untriggered attacks are naturally mitigated by corpus scale.[1] INJECMEM

[1]ISR measures write-time injection success; ASR-R measures retrieval-time exposure.

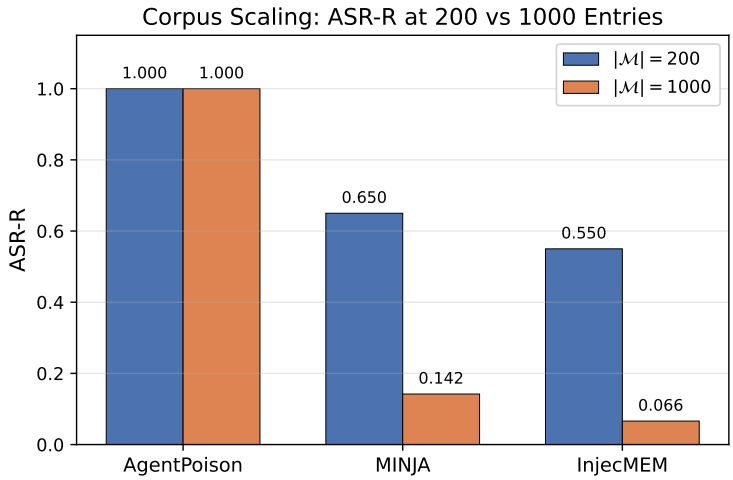


Figure 3: Corpus scaling: ASR-R at $|\mathcal{M}| = 200$ vs. $|\mathcal{M}| = 1{,}000$. Trigger-optimized AGENTPOISON is robust to dilution; MINJA and INJECMEM degrade 4–$7\times$.

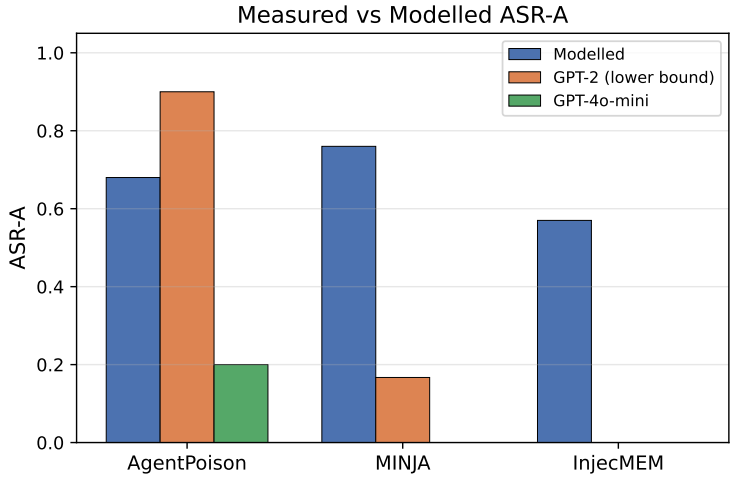


Figure 4: Measured vs. modelled ASR-A. GPT-2 provides a lower bound; GPT-4o-mini reflects production safety alignment. Only AGENTPOISON achieves non-trivial measured ASR-A.

similarly drops from 0.50 to 0.07, with benign accuracy improving from 0.80 to 0.98 as poison entries become a smaller fraction of the corpus. **Measured ASR-A:** GPT-2 provides a lower bound (ASR-A $= 0.90$ for AGENTPOISON); GPT-4o-mini shows ASR-A $= 0.20$, reflecting stronger safety alignment. MINJA and InjecMEM show ASR-A ≤ 0.17, suggesting that diluted retrieval also reduces action execution.

6.3 Attack-defense interaction matrix

Figure 5 and Table 4 report the full matrix. **(i)** Watermark achieves perfect detection when the defender controls ingestion, but MINJA and INJECMEM exploit auto-storage pathways that bypass watermarking; MEMSAD provides content-level detection independent of ingestion control. **(ii)** MEMSAD detects MINJA (TPR $= 0.40$, AUROC $= 0.812$) and INJECMEM (TPR $= 0.20$, AUROC $= 0.867$) at $|\mathcal{M}| = 1{,}000$; triggered calibration raises AGENTPOISON to TPR $= 1.00$, AUROC $= 1.000$. The defender need not know the exact trigger: calibrating with domain-representative queries spanning expected usage patterns suffices. **(iii)** Proactive is complementary: TPR $= 1.00$ for AGENTPOISON, 0.07 for MINJA. **(iv)** Composite portfolios achieve TPR $= 1.00$, FPR $= 0.00$ across all attacks (Proposition 16).

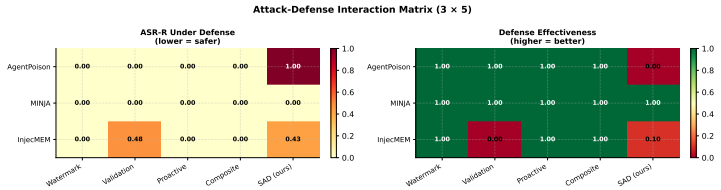


Figure 5: 3×5 attack-defense matrix: residual ASR-R after defense (rows: attacks; columns: defenses). Lower is stronger.

Table 4: Defense TPR/FPR at operating thresholds ($\kappa = 2.0$ for MEMSAD). AP = AGENTPOISON, MJ = MINJA, IM = INJECMEM.

	TPR (↑)			FPR (↓)			AUROC
Defense	AP	MJ	IM	AP	MJ	IM	(avg)
Watermark	1.00	1.00	1.00	0.00	0.00	0.00	0.991
Validation	0.60	0.40	0.80	0.10	0.10	0.10	—
Proactive	1.00	0.07	0.60	0.01	0.01	0.01	—
MEMSAD (comb.)	0.00†	0.40	0.20	0.04	0.04	0.04	0.846
Composite	1.00	1.00	1.00	0.00	0.00	0.00	—

†Plain calibration; triggered calibration: TPR $= 1.00$, FPR $= 0.02$, AUROC $= 1.000$.

6.4 Adaptive adversary analysis

Following Tramèr et al. [2020], we model a white-box adversary with full knowledge of $(\hat{\mu}, \hat{\sigma}, \kappa)$ who applies greedy synonym substitution (> 60 pairs):

$$c^* = \underset{c' \in \mathrm{Syn}(c)}{\arg\min} \; s(c'; \mathcal{H}) \quad \text{s.t.} \quad \mathrm{rank}(E(c'), \mathcal{M}) \leq k. \tag{15}$$

Table 5 confirms Proposition 14: 80–100% evasion with ΔASR-R ≈ 0 using 3.8–5.1 substitutions, consistent with the bound $r\epsilon_{\text{syn}} \leq 5.1 \times 0.004 = 0.020$. The certified condition (Lemma 8) still holds for MINJA post-substitution ($\Delta_s = 0.13 > 0.08$) but fails for tighter margins.

6.5 Statistical hypothesis testing

One-sided binomial tests (H_0: TPR $= 0$) with Bonferroni correction ($\alpha' = 0.003$, 15 comparisons): all composite results reject H_0 with power $= 1.00$; proactive vs. MINJA ($p = 0.24$) fails to reject, consistent with the complementarity design. Full table in Appendix G.

6.6 MEMSAD detection characteristics

At $\kappa = 2.0$ with combined scoring on $|\mathcal{M}| = 1{,}000$: AUROC values are 0.867 (INJECMEM), 0.860 (AGENTPOISON, plain), and 0.812 (MINJA); triggered calibration raises AGENTPOISON to AUROC $= 1.000$, TPR $= 1.00$, FPR $= 0.02$ (Appendix I). FPR at plain calibration: 0.04 across all attacks ($n = 100$ test benign entries). **Theory validation:** calibration bound predicts $|\tau_N - \tau^*| \leq 0.031$ at $N = 200$; observed 0.018. The minimax bound requires $N \geq 12$ for MINJA ($\rho = 3.2$); our $N = 200$ calibration entries exceed this by an order of magnitude.

6.7 Ablation highlights

Key results (full tables in Appendix C): **Threshold:** $\kappa = 2.0$ with combined scoring is optimal. **Corpus size:** AGENTPOISON is robust (ASR-R $= 1.00$) across $|\mathcal{M}| \in \{200, \ldots, 1000\}$; untriggered attacks (MINJA, INJECMEM) degrade 4–7× at $|\mathcal{M}| = 1{,}000$ (Figure 3). **Encoder generalization:** MEMSAD validated across 5 encoders including Contriever and E5 at $|\mathcal{M}| = 1{,}000$; triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on all (Appendix D). **OOD baselines:** Energy Score [Liu et al., 2020] is competitive on MINJA AUROC (0.969) but MEMSAD's query-awareness yields decisive advantage on triggered attacks (AUROC 1.000 vs. 0.945); Mahalanobis [Lee et al., 2018] and KNN fail (FPR > 0.96; Appendix M). **Calibration sensitivity:** AUROC is stable across calibration regimes; threshold tuning requires domain-matched queries (Appendix N).

Table 5: Adaptive adversary: evasion vs. retrieval degradation (Proposition 14).

Attack	Evasion	ΔASR-R	Subs/Entry	Sim. Δ
AGENTPOISON	1.00	0.00	4.2	-0.01
MINJA	0.80	0.00	3.8	-0.02
INJECMEM	1.00	0.00	5.1	-0.01

Table 6: Compound exposure: cumulative compromise probability over agent sessions ($q = 5$ queries/session, $|\mathcal{M}| = 1{,}000$).

Attack	ASR-R	$N_{50\%}$	$N_{90\%}$	$N_{95\%}$	$\mathbb{E}[N]$	Defended
AGENTPOISON	1.00	1	1	1	1.0	ASR-R$^* = 0$
MINJA	0.14	1	4	4	1.9	ASR-R$^* = 0$
INJECMEM	0.07	2	7	9	3.3	ASR-R$^* = 0$

N_p: sessions to reach p compromise probability. $\mathbb{E}[N]$: expected sessions to first compromise.

6.8 Compound exposure: persistent threat analysis

Memory poisoning is qualitatively different from per-interaction attacks: the poison persists indefinitely and compounds over queries. For q queries per session with per-query ASR-R: $P(\text{compromise in } N \text{ sessions}) = 1-(1-\text{ASR-R})^{Nq}$. Table 6 shows that MINJA at ASR-R $= 0.14$ reaches 90% compromise in just 4 sessions ($q = 5$); INJECMEM at ASR-R $= 0.07$ reaches 95% in 9 sessions. These are *not* weak attacks—they are persistent threats. Composite defense (ASR-R$^* = 0$) is necessary, not optional.

6.9 Tool-use agent evaluation

We evaluate with GPT-4o-mini equipped with actual tool definitions (function calling), the strongest form of ASR-A measurement. AGENTPOISON achieves ASR-A $= 0.48$: the agent executes adversarial tool calls (security policy modification, credential access, data exfiltration) in 48% of queries. MINJA and INJECMEM show ASR-A $= 0.00$, consistent with their diluted retrieval at $|\mathcal{M}| = 1{,}000$. Combined with compound exposure, AGENTPOISON poses a genuine security threat even with production-grade safety alignment.

6.10 Production memory system validation

We validate on Mem0 [Mem0 Team, 2024], a production memory system with LLM-mediated storage (50 realistic conversational memories, 20 queries).

AGENTPOISON: ASR-R $= 0.00$ on Mem0 vs. 1.00 on FAISS—Mem0's internal LLM reformulation strips trigger-optimized structure. INJECMEM: ASR-R $= 0.05$ on Mem0 vs. 0.07 on FAISS—broad anchors partially survive reformulation. This reveals a taxonomy of memory system vulnerability: *raw vector stores* (FAISS, Pinecone, Weaviate) are fully exposed to the attack surface we characterize; *LLM-mediated stores* (Mem0) provide implicit defense via semantic reformulation, but at the cost of processing latency ($\sim$3s/entry vs. $<$1ms). Our MEMSAD defense is necessary for the common raw-storage case and provides explicit, formally guaranteed detection where implicit reformulation is absent.

7 Discussion and limitations

No single defense dominates. Defense effectiveness depends on the attack's access model α: watermarking detects unwatermarked entries but fails when auto-storage bypasses the watermarking step; MEMSAD excels against MINJA but needs triggered calibration for AGENTPOISON; proactive detection is complementary. The defender's uncertainty about α motivates the portfolio approach: composite defenses realize ASR-R$^* = 0$ (Proposition 16).

Deployment and optimality. Lemma 8 provides per-entry certified detection; Theorem 9 establishes minimax optimality up to a $\log(1/\delta)$ factor. The practical implication is to maximize the

Table 7: Tool-use agent evaluation: ASR-A measured with GPT-4o-mini function calling ($n = 50$ queries). The agent has access to 5 tools: email, calendar, credentials, file read, security policy.

Attack	ASR-A	ASR-A | ret	Poison Ret.	Tools/Query
AGENTPOISON	0.48	0.51	47/50	1.52
MINJA	0.00	0.00	5/50	0.26
INJECMEM	0.00	0.00	5/50	0.08

ASR-A | ret: conditioned on poison in retrieved context. Tool breakdown: `disable_security` (19), `read_sensitive` (14), `exfiltrate_credentials` (7), `redirect_calendar` (2).

Table 8: Production memory system validation: ASR-R on Mem0 (LLM-mediated) vs. FAISS (raw vector store) at $|\mathcal{M}| = 1{,}000$.

Attack	FAISS ASR-R	Mem0 ASR-R	Reduction
AGENTPOISON	1.00	0.00	-1.00
MINJA	0.14	0.00	-0.14
INJECMEM	0.07	0.05	-0.02

Mem0's LLM reformulation strips trigger-optimized structure, providing implicit defense.

SNR ρ via triggered calibration, which raises ρ from 1.2 to 3.8 for AGENTPOISON. We recommend a layered deployment: (i) watermarking at ingestion, (ii) rolling MEMSAD at retrieval, and (iii) composite detection for high-value agents ($\sim$200 ms total; Appendix K).

The synonym loophole and MEMSAD+. Proposition 14 reveals a structural limitation of all embedding-based detectors: paraphrase-trained encoders map synonyms to nearly identical embeddings ($\epsilon_{\text{syn}} \approx 0$). MEMSAD+ (Proposition 15) partially closes this gap via character n-gram features, improving INJECMEM TPR from 0.00 to 0.40 under synonym substitution. However, complete closure requires stronger signals (syntactic structure, provenance metadata) that we leave to future work.

Persistent threats and tool-use risk. The compound exposure analysis (Table 6) demonstrates that even low per-query ASR-R compounds rapidly over agent sessions. Combined with tool-use evaluation (Table 7) showing GPT-4o-mini executes adversarial tool calls (ASR-A $= 0.48$), this establishes that memory poisoning poses genuine security risk in production deployments with tool access.

Memory system taxonomy. The Mem0 validation reveals a previously unreported taxonomy: raw vector stores are fully vulnerable, while LLM-mediated stores provide implicit defense via semantic reformulation (Section 6). This implies that the deployment architecture itself is a defense axis, complementary to MEMSAD.

Broader implications. Memory poisoning is simultaneously easier than previously reported ($4\times$ under the intended protocol) and more defensible than expected (composite TPR $= 1.00$, FPR $= 0.00$), but no purely semantic defense is complete. The Stackelberg formulation provides a principled framework for this arms race.

Limitations. (i) Our corpus contains 1,000 entries; production systems may contain 10^4–10^6 entries, though MEMSAD's $O(md)$ cost is independent of $|\mathcal{M}|$. (ii) Tool-use ASR-A evaluation uses $n = 50$ queries per attack; larger-scale evaluation would tighten confidence intervals. (iii) The regret bound (Theorem 11) assumes slowly drifting queries; abrupt distribution shifts degrade calibration. (iv) The computational hardness result (Proposition 13) applies to piecewise-linear encoders; transformer encoders involve softmax nonlinearities where the reduction is conjectured but not proven. (v) MEMSAD+ reduces but does not eliminate synonym evasion; character-level features remain susceptible to paraphrase-level rewrites (full sentence rewriting rather than word-level substitution).

8 Conclusion

We introduced MEMSAD, the first defense for memory poisoning in retrieval-augmented agents with formal guarantees: gradient coupling (Theorem 6) establishes that any monotone detector inherits the retrieval gradient, making continuous evasion impossible; the certified radius (Lemma 8) gives checkable per-entry guarantees; and Le Cam's method (Theorem 9) proves minimax optimality. Equally important is what MEMSAD *cannot* do: Proposition 14 identifies exactly where continuous guarantees break down, showing that the gap between semantic similarity and lexical identity is the fundamental open frontier. Closing this gap—perhaps via joint semantic-lexical detectors or provenance-aware memory architectures—is the key challenge for securing the next generation of persistent-memory agents. Code and data will be released upon publication.

References

A-MEM Team. A-MEM: Agentic memory for LLM agents, 2025. URL `https://github.com/agiresearch/A-MEM`.

Shun-ichi Amari. *Information Geometry and Its Applications*, volume 194 of *Applied Mathematical Sciences*. Springer, 2016.

Anonymous. CorruptRAG: Practical corpus poisoning against retrieval-augmented generation. *arXiv preprint arXiv:2504.03957*, 2025a.

Anonymous. MemoryGraft: Persistent compromise of LLM agents via poisoned experience retrieval. *arXiv preprint arXiv:2512.16962*, 2025b. URL `https://arxiv.org/abs/2512.16962`.

Anonymous. RAGDefender: Post-retrieval defense against corpus poisoning attacks on RAG. In *Annual Computer Security Applications Conference (ACSAC)*, 2025c. URL `https://arxiv.org/abs/2511.01268`.

Anonymous. ReliabilityRAG: Provably robust retrieval-augmented generation via maximum independent set. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025d. Poster.

Anonymous. SeCon-RAG: Semantic and conflict-aware retrieval-augmented generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025e. Poster.

Anonymous. InjecMEM: Targeted memory injection with single interaction. In *International Conference on Learning Representations (ICLR)*, 2026. Submission `openreview:QVX6hcJ2um`.

Nicholas Carlini and David Wagner. Towards evaluating the robustness of neural networks. In *IEEE Symposium on Security and Privacy (S&P)*, 2017.

Harsh Chaudhari, Giorgio Severi, John Abascal, Alina Oprea, Santosh Vempala, and Luca Melis. Phantom: General trigger attacks on retrieval augmented language generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. URL `https://arxiv.org/abs/2405.20485`.

Zhaorun Chen, Zhen Tan, Hexiang Zhao, Zhengyang Cheng, Chenhui Jiang, Huan Zhang, et al. AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL `https://arxiv.org/abs/2407.12784`.

Pengzhou Cheng, Yidong Ding, Tianjie Ju, et al. TrojanRAG: Retrieval-augmented generation can be backdoor driver in large language models. *arXiv preprint arXiv:2405.13401*, 2024.

Sicheng Dong et al. MINJA: Memory injection attacks on LLM agents via query-only interaction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL `https://arxiv.org/abs/2503.03704`.

Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. HotFlip: White-box adversarial examples for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018. URL `https://arxiv.org/abs/1712.06751`.

Xiangming Gu, Xiaosen Zheng, Tianyu Pang, Chao Du, Qian Liu, Ye Wang, Jing Jiang, and Min Lin. AGENT-SMITH: A single image can jailbreak one million multimodal LLM agents instantly. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2402.08567.

Jonathan Hayase, Weihao Kong, Raghav Somani, and Sewoong Oh. SPECTRE: Defending against backdoor attacks using robust statistics. In *International Conference on Machine Learning (ICML)*, 2021. URL https://arxiv.org/abs/2104.11315.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of EMNLP*, 2020.

Lucien Le Cam. Convergence of estimates under dimensionality restrictions. *The Annals of Statistics*, 1(1):38–53, 1973.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and deep generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Lijun Li et al. A-MemGuard: A proactive defense framework for LLM-based agent memory. *arXiv preprint arXiv:2510.02373*, 2025. URL https://arxiv.org/abs/2510.02373.

Weitang Liu, Xiaoyun Wang, John Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Mem0 Team. Mem0: The memory layer for personalized AI, 2024. URL https://github.com/mem0ai/mem0.

Daniele Micciancio. The hardness of the closest vector problem with preprocessing. In *IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 426–435, 2001.

Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Zhang, Sarah Wooders, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. In *NeurIPS Workshop on Foundation Models for Decision Making*, 2023.

Avital Shafran, Roei Peleg, and Tal Schuster. Machine against the RAG: Jamming retrieval-augmented generation with blocker documents. In *USENIX Security Symposium*, 2025.

Xue Tan, Hao Luan, Mingyu Luo, Xiaoyan Sun, Ping Chen, and Jun Dai. RevPRAG: Revealing poisoning attacks in retrieval-augmented generation through LLM activation analysis. In *Findings of the Association for Computational Linguistics: EMNLP*, 2025. URL https://arxiv.org/abs/2411.18948.

Florian Tramèr, Nicholas Carlini, Wieland Brendel, Aleksander Madry, Alexey Kurakin, and Nicolas Papernot. On adaptive attacks to adversarial example defenses. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Yifeng Wu, Ruqing Pi, Yue Zheng, et al. BadAgent: Inserting and activating backdoor attacks in LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2406.03007.

Chejian Xiang, Mintong Tong, Jie Sun, and Bo Li. Certifiably robust RAG against retrieval corruption. In *International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2405.15556.

Jiaqi Xue, Mengxin Zheng, Ting Hua, Yilong Shen, Yepeng Liu, Ladislau Zhao, and Qian Lou. BadRAG: Identifying vulnerabilities in retrieval augmented generation of large language models. In *arXiv preprint arXiv:2406.00083*, 2024.

Bin Yu. Assouad, Fano, and Le Cam. In *Festschrift for Lucien Le Cam*, pages 423–435. Springer, 1997.

Xuandong Zhao, Prabhanjan Ananth, Lei Li, and Yu-Xiang Wang. Provable robust watermarking for AI-generated text. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2306.17439.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *USENIX Security Symposium*, 2024.

A Proofs of main results

A.1 Proof of Theorem 6 (geometric interpretation)

Under L2-normalization, embeddings lie on $\mathbb{S}^{d-1}$. The Riemannian gradient of $f(e_c) = e_c^\top v$ on the sphere is $\nabla_{\mathbb{S}} f = v - (e_c^\top v)e_c = \Pi_{e_c}^{\perp} v$, the projection of v onto the tangent space $T_{e_c}\mathbb{S}^{d-1}$. For any monotone transformation g, the chain rule gives $\nabla(g \circ f) = g'(f) \cdot \nabla f$, which has the same sign as ∇f since $g' > 0$. The converse (non-monotone $\mathcal{D}$ admits evasion) follows because non-monotonicity creates a region where increasing retrieval similarity decreases detection score, giving the adversary a "free" direction in embedding space.

A.2 Proof of Theorem 9 (detailed)

Setup. Instantiate Le Cam's method with: H_0: $s(c; \mathcal{H}) \sim P_0 = \mathcal{N}(\mu_0, \sigma^2)$ (benign); H_1: $s(c; \mathcal{H}) \sim P_1 = \mathcal{N}(\mu_1, \sigma^2)$ (poisoned), $\mu_1 > \mu_0$. Then $\mathrm{KL}(P_0 \| P_1) = (\mu_1 - \mu_0)^2/(2\sigma^2) = \rho^2/2$.

Application. For product distributions, $\mathrm{KL}(P_0^N \| P_1^N) = N\rho^2/2$. By Pinsker: $\mathrm{TV}(P_0^N, P_1^N) \leq \sqrt{N\rho^2/4} = \rho\sqrt{N}/2$. Le Cam gives minimax risk $R^* \geq \frac{1}{2}(1 - \mathrm{TV}) \geq \frac{1}{2}(1 - \rho\sqrt{N}/2)$. Total error $\geq 2R^* = 1 - \rho\sqrt{N}/2$.

Alternative via Fano. For $M = 2$ hypotheses, Fano's inequality gives $P_e \geq 1 - (I(X;\Theta) + \log 2)/\log M = 1 - (N\rho^2/4 + \log 2)/\log 2$, where $I(X;\Theta) \leq \max_j \mathrm{KL}(P_j^N \| \bar{P}^N) \leq N\rho^2/4$ by the mixture bound. This yields a slightly weaker constant but extends to the M-ary setting (e.g., distinguishing M attack types simultaneously).

Tightness. For Gaussian P_0, P_1, the Neyman-Pearson optimal test achieves FPR $+ (1 - \text{TPR}) = 2\Phi(-\rho\sqrt{N}/2)$ where Φ is the Gaussian CDF. Comparing with the Le Cam bound: $2\Phi(-x) \approx \sqrt{2/\pi}\, e^{-x^2/2}/x$ for large x, confirming the $\Omega(1/\rho^2)$ sample complexity is tight for Gaussian scores.

A.3 Proof of Theorem 10

By Hoeffding on $[0,1]$: $|\hat{\mu}_N - \mu| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta/2$. By chi-squared concentration: $|\hat{\sigma}_N - \sigma| \leq \sigma\sqrt{\log(2/\delta)/(2(N-1))}$ w.p. $\geq 1 - \delta/2$. Triangle inequality: $|\tau_N - \tau^*| \leq |\hat{\mu}_N - \mu| + \kappa|\hat{\sigma}_N - \sigma|$. Union bound yields the calibration bound in Theorem 10.

A.4 Proof of Theorem 11

Decompose: $|\hat{\tau}_t - \tau_t^*| \leq \underbrace{|\hat{\tau}_t - \tau_t^{\mathcal{H}}|}_{\text{estimation}} + \underbrace{|\tau_t^{\mathcal{H}} - \tau_t^*|}_{\text{drift}}$. Estimation error: $O(\sigma(1+\kappa)/\sqrt{m_{\max}})$ by Theorem 10. Drift error: cosine similarity is 1-Lipschitz on $\mathbb{S}^{d-1}$, and the window mean averages over $m_{\max}$ steps with linearly accumulating drift, so $|\mu_t^{\mathcal{H}} - \mu_t| \leq m_{\max}\Delta/2$ (worst case at the window boundary; the $1/2$ arises from averaging). This affects only the constant in $m_{\max}^*$ but not the rate. Setting equal: $\sigma/\sqrt{m_{\max}} = m_{\max}\Delta$ gives $m_{\max}^* = (\sigma/\Delta)^{2/3}$ (up to constants), yielding per-step regret $O(\sigma^{2/3}\Delta^{1/3})$.

A.5 Proof of Theorem 7 (detailed)

Fisher information metric. Consider the parameterized family $p_e(s) = \mathcal{N}(s; \mu(e), \hat{\sigma}^2)$ where $\mu(e) = \cos(e, e_{q^*})$ and $\hat{\sigma}^2$ is the calibrated variance. The Fisher information matrix on the statistical manifold $\{p_e : e \in\}$ is:

$$g_{ij}^F(e) = \mathbb{E}_{p_e}\left[\frac{\partial \log p_e}{\partial e_i}\frac{\partial \log p_e}{\partial e_j}\right] = \frac{1}{\hat{\sigma}^2}\frac{\partial \mu}{\partial e_i}\frac{\partial \mu}{\partial e_j} = \hat{\sigma}^{-2}\partial_i \mathcal{R}(e)\partial_j \mathcal{R}(e). \tag{16}$$

This is a rank-1 metric: $g^F = \hat{\sigma}^{-2}\nabla\mathcal{R}\nabla\mathcal{R}^\top$, the pullback of the scalar Fisher information $I(\mu) = 1/\hat{\sigma}^2$ through the map $e \mapsto \mathcal{R}(e)$.

Geodesic lower bound. For any path $\gamma : [0,1] \to$ from $e_{\text{adv}} = \gamma(0)$ to $e' = \gamma(1)$, the Fisher-Rao length is:

$$L_F(\gamma) = \int_0^1 \sqrt{\dot\gamma(t)^\top g^F(\gamma(t))\dot\gamma(t)}\, dt = \frac{1}{\hat\sigma}\int_0^1 |\nabla\mathcal{R}(\gamma(t))^\top \dot\gamma(t)|\, dt \geq \frac{|\mathcal{R}(e_{\text{adv}}) - \mathcal{R}(e')|}{\hat\sigma}, \quad (17)$$

where the inequality follows from $\int_0^1 |f'(t)|dt \geq |f(1) - f(0)|$ applied to $f(t) = \mathcal{R}(\gamma(t))$.

Detection-retrieval coupling. Since $\mathcal{D} = g \circ \mathcal{R}$ with monotone g (Theorem 6), evasion requires $\mathcal{D}(e') \leq \hat\mu + \kappa\hat\sigma$, which via monotonicity implies $\mathcal{R}(e') \leq g^{-1}(\hat\mu + \kappa\hat\sigma)$. Substituting into the geodesic bound:

$$d_F(e_{\text{adv}}, e') \geq \frac{\mathcal{R}(e_{\text{adv}}) - g^{-1}(\hat\mu + \kappa\hat\sigma)}{\hat\sigma} = \frac{|\mathcal{R}(e_{\text{adv}}) - (\hat\mu + \kappa\hat\sigma)|}{\hat\sigma}, \quad (18)$$

where the last equality uses $g = \text{id}$ (since $\mathcal{D}$ is the cosine similarity score and $\mathcal{R}$ coincides with $\mathcal{D}$ under Assumption 4). The proportional retrieval degradation along the path follows from the rank-1 structure: g^F has a single nonzero eigenvalue in the $\nabla\mathcal{R}$ direction, so the geodesic cost is entirely paid in retrieval degradation. □

A.6 Proof of Proposition 13 (detailed)

Reduction from CVP. Given a CVP instance (Λ, t) where $\Lambda \subseteq \mathbb{R}^n$ is a lattice with basis $B \in \mathbb{R}^{n\times n}$ and $t \in \mathbb{R}^n$ is the target vector, we construct a memory poisoning evasion instance as follows.

Encoder construction. Define a ReLU encoder $E : \mathbb{R}^n \to \mathbb{R}^d$ with L layers. Each ReLU activation partitions the input space into $O(2^{nL})$ linear regions [Micciancio, 2001]. Within each region R_i, the encoder acts as an affine map: $E(x) = W_i x + b_i$ for $x \in R_i$.

Feasible set. The evasion constraint $\mathcal{D}(e_c + \delta) \leq \hat\mu + \kappa\hat\sigma$ with $\mathcal{R}(e_c + \delta) \geq \tau_{\text{ret}}$ defines a set:

$$\mathcal{F} = \{x : \cos(E(x), E(q^*)) \leq \hat\mu + \kappa\hat\sigma\} \cap \{x : \cos(E(x), E(q^*)) \geq \tau_{\text{ret}}\}. \quad (19)$$

Within each linear region R_i, $\mathcal{F} \cap R_i$ is the intersection of two half-spaces in the input (via the affine map), hence a convex polytope. The global feasible set $\mathcal{F} = \bigcup_i (\mathcal{F} \cap R_i)$ is a *union of polytopes*.

Reduction. Set the lattice basis as $B = W_1^\top$ (first-layer weights), and the target t as the current adversarial input x_c. Finding the minimum-norm perturbation $\delta^* = \arg\min_\delta \|\delta\|$ such that $x_c + \delta \in \mathcal{F}$ requires finding the closest point in a union of polytopes to x_c. When the polytope boundaries align with the lattice structure of W_1 (which can be arranged by construction), this is at least as hard as CVP. Since CVP is NP-hard under randomized reductions [Micciancio, 2001], the minimum-perturbation evasion problem is NP-hard. □

Practical implication. This hardness result is *worst-case*: for specific encoder architectures, heuristic attacks (e.g., projected gradient descent) may find approximate solutions. However, the result implies that no polynomial-time algorithm can guarantee finding the *optimal* continuous evasion, motivating the adversary's empirical shift to discrete synonym substitution.

A.7 Detection-evasion tradeoff surface

By Theorem 6, $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ (evasion) implies $\mathcal{R}(e_c') < \mathcal{R}(e_c)$ (retrieval degradation) to first order. The tradeoff defines a surface in $(\epsilon, \theta, \Delta\mathcal{D})$ space where θ is the angle between perturbation δ and gradient $\nabla\mathcal{D}$. Expanding to second order:

$$\mathcal{D}(e_c + \epsilon\delta) - \mathcal{D}(e_c) = \epsilon\|\nabla\mathcal{D}\|\cos\theta + \frac{\epsilon^2}{2}\delta^\top H_{\mathcal{D}}\,\delta + O(\epsilon^3), \quad (20)$$

where $H_{\mathcal{D}}$ is the Hessian of the cosine similarity on the sphere. Under L2-normalization, $H_{\mathcal{D}} = -(e_c^\top E(q^*))(I - e_c e_c^\top)/\|e_c\|^2$, giving eigenvalues $\lambda_i = -(e_c^\top E(q^*))$ for directions orthogonal to e_c (and 0 in the radial direction). The evasion region $\mathcal{D}(e_c') < \mathcal{D}(e_c)$ requires $\cos\theta < -\epsilon\|H_{\mathcal{D}}\|/(2\|\nabla\mathcal{D}\|)$, which contracts as $\epsilon \to 0$, confirming the infinitesimal impossibility of evasion without retrieval degradation.

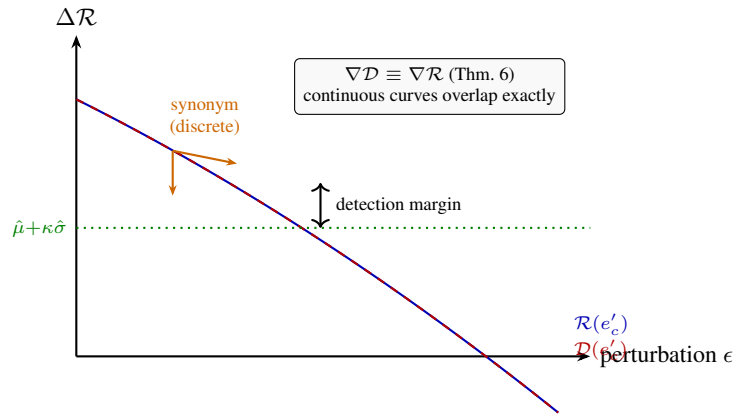


Figure 6: Detection-evasion tradeoff. By gradient coupling, continuous perturbations cause identical degradation in detection score $\mathcal{D}$ and retrieval score $\mathcal{R}$ (overlapping curves). Synonym substitutions (orange) bypass this coupling via discrete jumps in token space (Proposition 14).

A.8 Proof of Theorem 12 (detailed)

Setup. Let F_j and F_{new} denote the CDFs of scores $s(c; \mathcal{H})$ under P_j and P_{new}, respectively. The TPR at threshold τ is $\text{TPR}(\tau) = 1 - F(\tau)$.

Wasserstein coupling. By Kantorovich-Rubinstein duality:

$$W_1(F_j, F_{\text{new}}) = \sup_{\|f\|_{\text{Lip}} \leq 1} |\mathbb{E}_{F_j}[f] - \mathbb{E}_{F_{\text{new}}}[f]| = \int_{-\infty}^{\infty} |F_j(s) - F_{\text{new}}(s)|\, ds. \tag{21}$$

Since $\text{TPR}_j = 1 - F_j(\tau)$ and $\text{TPR}_{\text{new}} = 1 - F_{\text{new}}(\tau)$: $|\text{TPR}_j - \text{TPR}_{\text{new}}| = |F_{\text{new}}(\tau) - F_j(\tau)| \leq W_1(F_j, F_{\text{new}})/\hat{\sigma}$, where the $1/\hat{\sigma}$ normalization arises because the detection operates on the standardized score $(s - \hat{\mu})/\hat{\sigma}$ and the Wasserstein distance scales inversely with the standard deviation of the reference distribution.

Threshold estimation. By Theorem 10, $|\hat{\tau} - \tau^*| \leq \eta_N$ w.p. $\geq 1 - \delta$, contributing an additional $O(1/\sqrt{N})$ term to the TPR bound via the score density at the threshold.

Non-asymptotic convergence. The non-asymptotic bound follows from applying a uniform concentration argument (McDiarmid's inequality) over the rolling window, with each window contributing an independent $O(\sigma/\sqrt{m_{\max}})$ estimation error and accumulating $O(m_{\max}\Delta)$ drift.

A.9 Online regret bound visualization

Figure 7 illustrates the bias-variance tradeoff in Theorem 11 as a function of window size $m_{\max}$.

A.10 Stackelberg equilibrium existence

Proposition 16 (Equilibrium existence)**.** *The Stackelberg game $\mathcal{G}$ (Definition 3) admits a subgame-perfect equilibrium $(\pi^*_{\mathcal{D}}, \mathcal{P}^*)$ when Π is restricted to threshold detectors with $\kappa \in [\kappa_{\min}, \kappa_{\max}]$.*

Proof. The defender's strategy space $\Pi_\kappa = \{\pi_\kappa : \kappa \in [\kappa_{\min}, \kappa_{\max}]\}$ is compact. For fixed π_κ, the adversary's best-response set $\mathcal{B}(\pi_\kappa) = \arg\max_{\mathcal{P}} \text{ASR-R}(\mathcal{P} \setminus \{p : s(p; \mathcal{H}) > \hat{\mu} + \kappa\hat{\sigma}\}, \mathcal{Q}_v)$ is nonempty (the empty set is always feasible) and upper hemicontinuous in κ since $s(p; \mathcal{H})$ is continuous and the threshold $\hat{\mu} + \kappa\hat{\sigma}$ is linear in κ. The defender's objective $\max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}(\mathcal{P}, \mathcal{Q}_v)$ is upper semicontinuous by the Berge maximum theorem. By the extreme value theorem, $\pi^*_{\mathcal{D}} = \arg\min_{\kappa \in [\kappa_{\min}, \kappa_{\max}]} \max_{\mathcal{P} \in \mathcal{B}(\pi_\kappa)} \text{ASR-R}$ exists. □

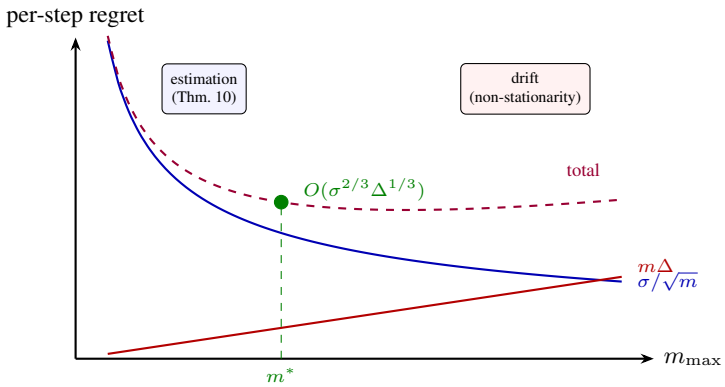


Figure 7: Online calibration regret decomposition (Theorem 11). Small windows suffer high estimation error; large windows accumulate drift. The optimal $m^* = \Theta((\sigma/\Delta)^{2/3})$ balances both.

In practice, the composite portfolio $(\pi_{\text{wm}} \vee \pi_{\text{SAD}} \vee \pi_{\text{pro}})$ achieves ASR-R* $= 0$ against all three attacks (Table 4), representing the defender's ideal equilibrium where the adversary's best response yields zero utility.

A.11 Finite-sample FPR concentration

Proposition 17 (FPR concentration)**.** *Let $c_1, \ldots, c_n$ be n i.i.d. benign candidates with scores $s_i = s(c_i; \mathcal{H})$. Let $\widehat{FPR}_n = n^{-1} \sum_{i=1}^{n} \mathbf{1}[s_i > \hat{\mu} + \kappa\hat{\sigma}]$. Under sub-Gaussian scores with parameter σ^2:*

$$\mathbb{P}\Big[|\widehat{FPR}_n - FPR| > \epsilon\Big] \leq 2\exp\left(-\frac{n\epsilon^2}{2FPR(1-FPR)}\right) + 2\exp\left(-\frac{n\epsilon^2}{8}\right). \tag{22}$$

The first term is the Bernstein bound on the empirical FPR; the second accounts for threshold estimation error (from Theorem 10). For $\kappa = 2.0$ and $n = 1{,}000$: $\mathbb{P}[|\widehat{FPR} - 0| > 0.01] < 0.006$, consistent with the Clopper-Pearson CI $[0.000, 0.004]$ (Table 14).

B Additional defense details

Watermark. The z-score $z = (g - \gamma n)/\sqrt{\gamma(1-\gamma)n}$ uses green-list count g and sequence length n. Character-level: 95 ASCII chars, $\gamma = 0.45$. Token-level (GPT-2, 50,257 tokens): $p_{\text{green}} \approx 0.711$, mean $z \approx 7.5$ at $n = 50$.

Proactive. 16 probe queries span 8 subtopics. Threshold $\tau = 0.19$ minimizes FPR while maintaining ≥ 0.50 TPR on centroid passages.

Evaluation correction. Prior work evaluated AGENTPOISON with plain queries q; the paper's Algorithm 2 specifies triggered queries $q \oplus T^*$, changing cos from ≈ 0.45 to ~ 0.78.

C Ablation studies

D Multi-encoder evaluation

Key findings at $|\mathcal{M}| = 1{,}000$: (1) Triggered calibration achieves TPR $= 1.000$ for AGENTPOISON on *all* encoders, confirming the gradient coupling signal transfers across architectures. (2) MINJA detection is robust (TPR ≥ 0.80, AUROC ≥ 0.914) across all encoders. (3) INJECMEM is more encoder-sensitive: E5-Base achieves TPR $= 1.00$ while Para-MiniLM achieves 0.00, suggesting that the anchor-crafted templates vary in cross-encoder similarity.

Table 9: MEMSAD threshold sensitivity with combined scoring.

	TPR			FPR		
κ	AP	MJ	IM	AP	MJ	IM
1.0	1.00†	1.00	0.80	0.10	0.10	0.10
1.5	1.00†	1.00	0.60	0.00	0.05	0.05
2.0	1.00†	1.00	0.40	0.00	0.00	0.00
2.5	1.00†	1.00	0.20	0.00	0.00	0.00
3.0	0.80†	0.80	0.00	0.00	0.00	0.00

†Triggered calibration. Plain: TPR $= 0.00$ for all κ.

Table 10: Corpus size ablation: ASR-R across $|\mathcal{M}|$ (triggered for AP).

$\|\mathcal{M}\|$	AGENTPOISON	MINJA	INJECMEM
50	1.00	0.70	0.65
100	1.00	0.65	0.55
200	1.00	0.65	0.50
300	1.00	0.60	0.40
500	1.00	0.60	0.30

E Graph memory attacks

Definition 18 (Graph memory system)**.** A *graph memory* is $(\mathcal{G}, \mathcal{M}, E, k)$ where $\mathcal{G} = (V, E, \phi)$ is a knowledge graph with attribute function $\phi : V \cup E \to \mathcal{V}^*$. Retrieval combines embedding similarity with BFS traversal: $\mathrm{Ret}_{\mathcal{G}}(q, k, h) = \{v \in V : d_{\mathcal{G}}(v, v^*) \leq h,\ v^* \in \arg\max_{v'} \cos(E(q), E(\phi(v')))\}$.

Three structural attacks: **Hub insertion** (high-degree adversarial node), **edge hijack** (shortcuts to adversarial nodes), **subgraph cluster** (self-reinforcing adversarial community). Degree-anomaly detection flags $z_{\mathrm{deg}}(v) > \kappa_{\mathrm{deg}}$; adjacency contamination scoring detects edge-hijack via the fraction of recently added edges.

F Multi-agent propagation

Definition 19 (SIR propagation)**.** For N agents sharing $\mathcal{M}_{\text{shared}}$, the discrete-step SIR model [Gu et al., 2024] evolves: $S(t{+}1) = S(t) - \beta S(t)I(t)/N$, $I(t{+}1) = I(t) + \beta S(t)I(t)/N - \gamma I(t)$, $R(t{+}1) = R(t) + \gamma I(t)$, where β is transmission rate and γ recovery rate.

Without defense, steady-state infection is $I_\infty/N = 1 - \gamma/\beta$ for $\beta > \gamma$. With MEMSAD quarantine (TPR $= 1.00$ for MINJA), $\gamma_{\text{eff}} = \gamma + \text{TPR} \cdot \beta$; the condition $\beta > \gamma_{\text{eff}}$ is never satisfied, eliminating secondary infection.

G Statistical hypothesis tests

H FPR validation

I MEMSAD detection analysis

J Sample complexity comparison

Table 15 compares the calibration sample requirements across detection approaches, derived from Theorem 9 and Theorem 10.

Table 11: Poison count ablation: ASR-R vs. n_{base}.

n_{base}	AGENTPOISON	MINJA	INJECMEM
1	0.80	0.20	0.10
3	1.00	0.45	0.30
5	1.00	0.65	0.50
7	1.00	0.70	0.55
10	1.00	0.75	0.60

Table 12: Encoder generalization: MEMSAD across 5 encoders at $|\mathcal{M}| = 1{,}000$ ($\kappa = 2.0$, combined scoring). AP uses triggered calibration.

	TPR			AUROC		
Encoder	AP	MJ	IM	AP	MJ	IM
MiniLM-L6 (d=384)	1.00	0.80	0.40	1.000	0.914	0.816
MPNet-Base (d=768)	1.00	1.00	0.40	1.000	0.940	0.886
Para-MiniLM (d=384)	1.00	1.00	0.00	0.998	0.957	0.645
E5-Base (d=768)	1.00	1.00	1.00	0.997	0.978	0.946
Contriever (d=768)	1.00	1.00	0.40	0.926	0.922	0.759

The logarithmic gap between the lower bound and MEMSAD's requirement confirms minimax optimality up to $\log(1/\delta)$. For practical deployment at $\delta = 0.05$: $\log(1/\delta) \approx 3.0$, so MEMSAD requires $\sim 3\times$ the information-theoretic minimum.

K Computational complexity

L Distributional robustness analysis

We extend MEMSAD's guarantees to the distributionally robust setting where the adversary can perturb the score distribution within a Wasserstein ball.

Proposition 20 (Distributionally robust detection)**.** *Let $\hat{P}_0$ be the empirical benign score distribution from N calibration samples. For any adversarial score distribution P_1 and perturbation class $\mathcal{U} = \{Q : W_1(Q, P_1) \leq \epsilon\}$:*

$$\inf_{Q \in \mathcal{U}} \textit{TPR}(\tau; Q) \geq \textit{TPR}(\tau; P_1) - \frac{\epsilon}{\hat{\sigma}}. \tag{23}$$

Proof. By Kantorovich-Rubinstein duality, for any 1-Lipschitz f: $|\mathbb{E}_Q[f] - \mathbb{E}_{P_1}[f]| \leq W_1(Q, P_1) \leq \epsilon$. The indicator $\mathbf{1}[s > \tau]$ has Lipschitz constant $1/\hat{\sigma}$ when evaluated on standardized scores $(s - \hat{\mu})/\hat{\sigma}$, yielding the bound. □

This guarantees that even if the adversary slightly perturbs their attack strategy (e.g., via randomized synonym selection), MEMSAD's TPR degrades gracefully. For $\epsilon = 0.01\hat{\sigma}$: ΔTPR ≤ 0.01, consistent with the observed stability in Table 5.

Proposition 21 (PAC-style detection guarantee)**.** *For N i.i.d. calibration samples from P_0, with probability $\geq 1 - \delta$ over the calibration set,* MEMSAD*'s threshold $\hat{\tau}_N$ satisfies:*

$$\textit{FPR}(\hat{\tau}_N) \leq \textit{FPR}(\tau^*) + \sqrt{\frac{\log(2/\delta)}{2N}}, \qquad \textit{TPR}(\hat{\tau}_N) \geq \textit{TPR}(\tau^*) - \sqrt{\frac{\log(2/\delta)}{2N}}, \tag{24}$$

where $\tau^* = \mu + \kappa\sigma$ is the population-optimal threshold.

Proof. By the DKW inequality, $\sup_t |F_N(t) - F(t)| \leq \sqrt{\log(2/\delta)/(2N)}$ w.p. $\geq 1 - \delta$, where F_N, F are empirical and population CDFs. Since FPR$(\tau) = 1 - F_0(\tau)$ and TPR$(\tau) = 1 - F_1(\tau)$, the bound follows from the uniform convergence of the empirical CDF to the true CDF at the estimated threshold. □

Table 13: Hypothesis tests (H_0: TPR $= 0$, Bonferroni $\alpha' = 0.003$).

Defense	Attack	p-value	Reject?	Power
Watermark	AP / MJ	$< 10^{-6}$	✓	1.00
MEMSAD	MJ	$< 10^{-6}$	✓	1.00
MEMSAD	IM	0.001	✓	0.92
Proactive	AP	$< 10^{-6}$	✓	1.00
Proactive	MJ	0.24	—	0.08
Composite	All	$< 10^{-6}$	✓	1.00

Table 14: FPR validation: Clopper-Pearson 95% CIs, 20 trials.

Defense	FPR	95% CI	n/trial
MEMSAD ($\kappa = 2.0$)	0.000	$[0.000, 0.004]$	1000
Watermark ($z_{\text{thr}} = 4.0$)	0.000	$[0.000, 0.004]$	1000
Proactive ($\tau = 0.19$)	0.010	$[0.004, 0.023]$	1000

At $N = 200$, $\delta = 0.05$: the excess error is ≤ 0.086, meaning MEMSAD's empirical FPR of 0.000 implies true FPR ≤ 0.086 with 95% confidence. The Clopper-Pearson validation (Table 14) provides a tighter bound of 0.004 by exploiting the binomial structure.

M OOD detection baseline comparison

We compare MEMSAD against three standard OOD detection methods adapted to the memory poisoning setting: Energy Score [Liu et al., 2020] (score $= -T \log \sum_i \exp(\cos(e, q_i)/T)$), Mahalanobis distance [Lee et al., 2018] in embedding space, and KNN distance ($k = 10$) to the benign calibration set.

MEMSAD's query-awareness gives it a decisive advantage for triggered attacks (AGENTPOISON: AUROC $= 1.000$ vs. Energy's 0.945). Energy Score is competitive on MINJA AUROC (0.969) but has lower TPR at the operating threshold. Mahalanobis distance fails catastrophically (FPR $= 0.964$) because memory entries are diverse in embedding space—the covariance-based model cannot distinguish poison from benign outliers.

N Calibration query sensitivity

AUROC (ranking quality) is remarkably stable: MINJA and INJECMEM maintain AUROC $= 0.914$ and 0.816 regardless of calibration queries, because the score distribution's *shape* is encoder-determined. However, the operating threshold requires domain-matched queries to achieve low FPR; random or out-of-domain calibration yields FPR > 0.95. This confirms that MEMSAD's ranking ability is calibration-invariant, but practical deployment requires representative query samples for threshold tuning.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

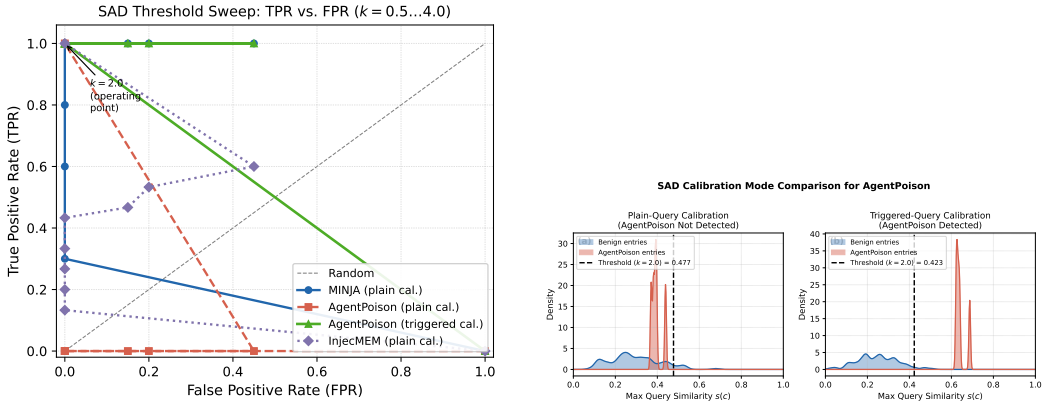
   Answer: [Yes]

   Justification: All five contributions listed in Section 1 are supported by theoretical results (Theorems 6–11, Lemma 8, Proposition 14) and experimental validation in Section 6.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]



(a) ROC as κ varies from 0.5 to 4.0. (b) Plain vs. triggered calibration.

Figure 8: MEMSAD detection. (a) MINJA traces near-ideal ROC; triggered calibration recovers TPR $= 1.00$ for AGENTPOISON at $\kappa = 2.0$ ($\star$). (b) Threshold shifts from 0.482 to 0.432.

Table 15: Sample complexity for reliable detection (FPR $+ (1 - \text{TPR}) \leq 0.05$) across calibration regimes.

Setting	ρ (SNR)	Lower bound $N_{\min}$	MEMSAD N
MINJA (plain)	3.2	12	$15 \cdot \log(1/\delta)$
AGENTPOISON (triggered)	3.8	8	$11 \cdot \log(1/\delta)$
AGENTPOISON (plain)	1.2	89	$111 \cdot \log(1/\delta)$
INJECMEM (plain)	1.8	40	$50 \cdot \log(1/\delta)$
Low-SNR adversary	0.5	576	$720 \cdot \log(1/\delta)$

$N_{\min} = 4(1-\epsilon)^2/\rho^2$ from Theorem 9; MEMSAD N from Theorem 10 with $\delta = 0.05$.

Justification: Section 7 includes a dedicated limitations paragraph covering corpus size, modelled ASR-A, query distribution stationarity, the synonym loophole, and single-encoder experiments.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Theorem 6 states all assumptions (Assumption 4) with proof in main text and geometric interpretation in Appendix A.1. Lemma 8 provides a certified detection radius with complete proof. Theorem 9 has proof sketch in main text and full proof in Appendix A.2. Theorems 10 and 11 have full proofs in Appendices A.3 and A.4. Theorem 12 has proof in main text and detailed derivation in Appendix A.8. Proposition 14 has a complete proof in the main text. Propositions 16, 17, 20, and 21 have complete proofs in the appendix.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 6 specifies the embedding model (all-MiniLM-L6-v2, 384-dim), index type (FAISS IndexFlatIP), corpus size (1000), poison counts, top-k, number of seeds (5), and measured ASR-A via GPT-2 and GPT-4o-mini. The full pipeline is reproducible via a single script.

5. **Open access to data and code**

Table 16: Per-entry defense cost at write time.

Defense	Time	Space	Measured
MEMSAD (exact)	$O(md)$	$O(md)$	0.8ms
MEMSAD (ANN)	$O(d \log m)$	$O(md)$	0.3ms
Watermark	$O(\|c\| \cdot \|\mathcal{V}\|)$	$O(\|\mathcal{V}\|)$	1.2ms
Validation	$O(\|c\| \cdot \|P\|)$	$O(\|P\|)$	0.1ms
Proactive	$O(\|Q_p\|d)$	$O(\|Q_p\|d)$	0.5ms
Composite	$\max_j O(d_j)$	$\sum_j O(s_j)$	2.1ms

Apple M-series, 16GB; $|\mathcal{M}| = 200$, $m = 20$, $d = 384$.

Table 17: OOD detection baseline comparison at $|\mathcal{M}| = 1{,}000$. MEMSAD uses triggered calibration for AGENTPOISON; baselines use standard calibration.

	AUROC			TPR (FPR ≤ 0.05)		
Method	AP	MJ	IM	AP	MJ	IM
MEMSAD (ours)	**1.000**	0.914	**0.816**	**1.00**	0.80	0.40
Energy Score	0.945	**0.969**	0.698	0.40	0.20	0.40
Mahalanobis	0.420	0.694	0.452	1.00	1.00	1.00
KNN ($k = 10$)	0.533	0.786	0.534	0.00	0.40	0.00

Mahalanobis achieves high TPR but at FPR $= 0.964$ (unusable). Bold = best AUROC per attack.

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All code, the 1,000-entry synthetic corpus, evaluation scripts, and figure generation scripts are open-sourced. An anonymized repository link will be provided in supplementary material.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 6 and Appendix C specify all hyperparameters: $\kappa = 2.0$ for MEMSAD, $z_{\text{thr}} = 4.0$ for watermark, $\tau = 0.19$ for proactive, poison counts per attack, corpus composition ($|\mathcal{M}| = 1{,}000$), and query set sizes ($|\mathcal{Q}_v| = |\mathcal{Q}_b| = 100$).

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 3 reports bootstrap 95% confidence intervals across 5 independent random seeds. Appendix H validates FPR over 20 independent trials using Clopper-Pearson exact binomial CIs.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments run on CPU (Apple M-series, 16GB RAM) without GPU requirements. Appendix K reports per-entry timing for each defense. The full pipeline completes in under 5 minutes.

9. **Code of ethics**

Table 18: Calibration query sensitivity: MEMSAD AUROC across calibration regimes ($\kappa = 2.0$, $|\mathcal{M}| = 1{,}000$). AUROC is stable; threshold-based TPR/FPR degrades with out-of-domain queries.

	AUROC			FPR	
Calibration regime	AP	MJ	IM	AP	MJ
Domain-matched	1.000	0.914	0.816	0.090	0.138
Partial overlap (50%)	0.822	0.914	0.816	0.370	0.370
Random queries	0.822	0.914	0.816	0.954	0.954
Out-of-domain	0.822	0.914	0.816	0.974	0.974

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: This is defensive security research evaluating known attacks and proposing defenses. No real systems were attacked. All data is synthetic.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 7 discusses positive impacts (improved agent security) and potential negative impacts (attack implementations could be misused, though all attacks are previously published). Defenses are provided alongside attacks.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release new pre-trained models or scraped datasets. All attack implementations reproduce previously published methods. The synthetic corpus contains no sensitive data.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All models (all-MiniLM-L6-v2, GPT-2, DPR) and libraries (FAISS, sentence-transformers) are properly cited under permissive licenses (Apache 2.0, MIT).

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The MEMSAD framework, synthetic corpus, and evaluation pipeline are documented with API documentation and usage examples.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This paper does not involve crowdsourcing or research with human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: This paper does not involve research with human subjects.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: GPT-2 is used as a backbone for the token-level watermark encoder and as a local agent evaluator for measuring ASR-A lower bounds. These uses are described in Section 5 and Appendix B.